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# Week 5 : Assignment 5

**The due date for submitting this assignment has passed.**

**Due on 2025-02-26, 23:59 IST.**

## Assignment submitted on 2025-02-26, 21:50 IST

1) Which of the following is the correct way to add data with key as `CS102` and value as `"Database Management Systems"` to a dictionary named `courses`? ***1 point***

- `courses["CS101"] = ("Database Management Systems")`
- `courses["CS102"]["Database Management Systems"]`
- **`courses["CS102"] = "Database Management Systems"`**
- `courses["CS101"] = "Database Management Systems"`

Yes, the answer is correct.
Score: 1
Accepted Answers:
`courses["CS102"] = "Database Management Systems"`

2) What is the probability of Monty not opening the door with goat, given the hypothesis that you initially chose the door which has car ? ***1 point***

- **0**
- 0.5
- 1
- 0.33

Yes, the answer is correct.
Score: 1
Accepted Answers:
*0*

3) What should be replaced with `?` in line 10, so that there is high chance that `final_choice` is equal to 2 ? ***1 point***

```
import random
initial_choice = random.randint(0, 2)
doors = ['goat', 'goat', 'car']

for i in range(3):
    if i != initial_choice and doors[i] != 'car':
        monty_opens = i
        break

if ? :
    for i in range(3):
        if i != initial_choice and i != monty_opens:
            final_choice = i
            break
else:
    final_choice = initial_choice
```

- **2**
- `2 * *(1024)` % 2 != 0
- **`2 * (2 * *(89) + 7)` % 2 == 0**

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**Week 6 ()**
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$(2 * *(90))$ % len(doors) != 1

Yes, the answer is correct.
Score: 1
Accepted Answers:
*2*
$2 * (2 * *(89) + 7)$ *% 2 == 0*

4) Given that you have a sorted list of 2048 elemets,what is the maximum number of comparisons to search such an element using linear search ? Can binary search do it in less number of comparisons ? ***1 point***

- Yes, binary search can do it more efficiently, linear search takes 1024 comparisons
- Yes, binary search can do it more efficiently, linear search takes 2048 comparisons
- No, binary search may/may not do it more efficiently, linear search takes 1024 comparisons
- No, binary search will take more comparisons, linear search takes 2048 comparisons

Yes, the answer is correct.
Score: 1
Accepted Answers:
*Yes, binary search can do it more efficiently, linear search takes 2048 comparisons*

5) What is the primary use of a file with a .wav or .wave extension? ***1 point***

- Storing video recordings
- Storing log files
- Storing waves of files
- Storing audio recordings

Yes, the answer is correct.
Score: 1
Accepted Answers:
*Storing audio recordings*

6) What does this program print in the end ? ***1 point***

```
import random
n = 10
counter = 0
for i in range(n):
  choices = ["rock", "paper", "scissors"]
  choice1 = random.choice(choices)
  choice2 = random.choice(choices)
  if choice1 != choice2:
    counter += 1
print(counter/n)
```

- Fraction of throws where both players showed different symbol(rock/paper/scissors)
- Fraction of throws where both players showed rock.
- Fraction of throws where both players showed same symbol(rock/paper/scissors)
- Fraction of throws where both players showed paper.

No, the answer is incorrect.
Score: 0
Accepted Answers:
*Fraction of throws where both players showed different symbol(rock/paper/scissors)*

7) In binary search the list gets divided in every iteration to find the element. ***1 point***

- True
- False

Yes, the answer is correct.
Score: 1
Accepted Answers:
*True*

8) Which of the following are requirements for binary search to work correctly? ***1 point***

- The list can be unsorted.
- The list must contain only positive integers.
- The list must be sorted.
- The list must have no duplicate values.

Yes, the answer is correct.
Score: 1
Accepted Answers:
*The list must be sorted.*

**Text Transcripts ()**

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9) Given an array [4, 2, 7, 1, 3], what will be the array after the third pass of Bubble Sort? ***1 point***

- $[1, 3, 2, 4, 7]$
- $[2, 4, 1, 3, 7]$
- $[1, 2, 3, 4, 7]$
- $[2, 1, 3, 4, 7]$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$[1, 2, 3, 4, 7]$

10) How many swaps are performed in iteration 3 for bubble sorting list [5,4,3,2,1]? Enter 0 if the list is sorted in less than 3 iterations.

2

Yes, the answer is correct.
Score: 1

Accepted Answers:
*(Type: Numeric) 2*

***1 point***